Solution of Nonlinear Equations

Bisection method

Let $f(x)$ be a continuous function and defined on $[a,b]$, let $f(a)$ and $f(b)$ have opposite signs $f(a)\cdot f(b)<0$. Then there exist on p such that $f(p)=0$ and $p\in(a,b)$.

The steps of solve bisection method

١. نجد قيم $p=\dfrac{a+b}{2}$

٢. نحسب قيمة $f(p)$ و هناك ثلاث أحتمالات

أ. أذا كانت $f(p)=0$ فان p هو الجذر المطلوب.

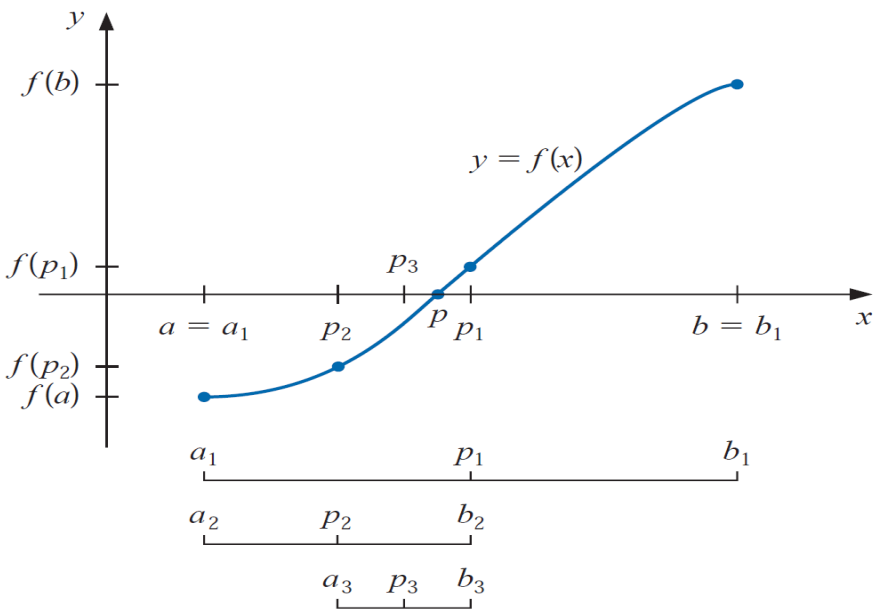
ب. أذا كانت $f(a)f(p)<0$ فان للمعادلة جذر يقع ضمن الفترة $[a,p]$.

ج. أذا كانت $f(b)f(p)<0$ فان للمعادلة جذر يقع ضمن الفترة $[p,b]$.

٣. تكرر الخطوة (٢،١) الى أن نصل للجذر المطلوب ونسبة خطأ لاتتجاوز ε.

* شرط التوقف: يمكن أستخدام الصيغ الأتية لغرض التوقف عن التكرار.

1. $|p_n - p_{n-1}| \le \varepsilon$
2. $|f(p_n)| \le \varepsilon$



Ex. (1):- Find the root of the equation $f\ (x) = x^3, x \in [-1,1]$

Solu. $f(-1) = -1, f(1) = 1$

$$p = \frac{a+b}{2} = \frac{1-1}{2} = 0$$

$$f(0) = 0$$

n	a	b	p	$f(p)$
1	-1	1	0	0

$P = 0$ is a root

Ex. (2):- Find the root of the equation $f\ (x) = 2x - 1, x \in [0,2]$

Solu. $f(0) = -1\ f(2) = 3$

$$p = \frac{a+b}{2} = \frac{2+0}{2} = 1$$

n	a	b	p	$f(p)$
1	0	2	1	1
2	0	1	0.5	0

$$f(1) = 2 - 1 = 1$$

1. $f(0)f(1) < 0$
2. $f(2)f(1) > 0$

$$p = \frac{1+0}{2} = 0.5$$

$$f(0.5) = 2 * 0.5 - 1 = 0$$

$p = 0.5$ is a root

Ex. (3):- Find the approximate positive value of the root of the equation $f\ (x) = cosx - x^2, x \in [0,1]$ by using bisection method with error $\varepsilon = 0.01$

Solu. first, we must determine a & b

$f(x) = 0 \Rightarrow \cos x = x^2$

$a = 0, b = 1$

$$f\ (0) = cos0 - 0 = 1$$

$$f(1) = cos(1 * \frac{180}{\pi}) - 1 = -0.46$$

$$f(0)f(1) < 0$$

n	A	b	p	f(p)
1	0	1	0.5	0.628
2	0.5	1	0.75	0.169
3	0.75	1	0.875	-0.125
4	0.75	0.875	0.813	0.026
5	0.813	0.875	0.844	-0.048
6	0.813	0.844	0.829	-0.011
7	0.813	0.829	0.821	0.007

* $p = \frac{a+b}{2} = \frac{0+1}{2} = 0.5$

$$f(0.5) = cos(0.5 * \frac{180}{\pi}) - (0.5)^2 = 0.628$$

$$f(1)f(0.5) < 0$$

* $p = \frac{0.5+1}{2} = 0.75$

$$f(0.75) = cos(0.75 * \frac{180}{\pi}) - (0.75)^2 = 0.169$$

$$f(1)f(0.75) < 0$$

* $p = \frac{0.75+1}{2} = 0.875$

$$f(0.875) = cos(0.875 * \frac{180}{\pi}) - (0.875)^2 = -0.125$$

$$f(0.75)f(0.875) < 0$$

* $p = \frac{0.75+0.875}{2} = 0.813$

$$f(0.813) = cos(0.813 * \frac{180}{\pi}) - (0.813)^2 = 0.026$$

$$f(0.813)f(0.875) < 0$$

* $p = \frac{0.813+0.875}{2} = 0.844$

$$f(0.844) = cos(0.844 * \frac{180}{\pi}) - (0.844)^2 = -0.048$$

$$f(0.813)f(0.844) < 0$$

* $p = \frac{0.813 + 0.844}{2} = 0.829$

$$f(0.829) = cos(0.829 * \frac{180}{\pi}) - (0.829)^2 = -0.011$$

$$f(0.813)f(0.829) < 0$$

* $p = \frac{0.813 + 0.829}{2} = 0.821$

$$f(0.821) = cos(0.821 * \frac{180}{\pi}) - (0.821)^2 = 0.007$$

$$|0.821 - 0.829| = 0.008 < 0.01$$

$\therefore$ the root is $p = 0.821$

Theorem: Let $f \in c[a,b]$ and suppose $f(a)f(b) < 0$, the bisection method generates a sequence $\{p_n\}_{n=1}^{\infty} \to p$ with property $|p_n - p| \le \frac{b-a}{2^n}, n \ge 1$

Problem: Determine approximates how many iterations are needed (necessary) to solve $f(x) = cosx - x2 = 0, x \in [0,1]$ with accuracy $\varepsilon = 0.01$

$|p_n - p| \le \frac{b-a}{2^n}$, $|p_n - p| \le \varepsilon$

$\Rightarrow \quad \frac{b-a}{2^n} < \varepsilon \Rightarrow \frac{1-0}{2^n} = \frac{1}{2^n} = 2^{-n} < 10^{-2}$

$\Rightarrow \log_{10} 2^{-n} < \log_{10} 10^{-2}$

$\Rightarrow -n \log_{10} 2 < -2 \log_{10} 10$

$\Rightarrow -n \log_{10} 2 < -2$

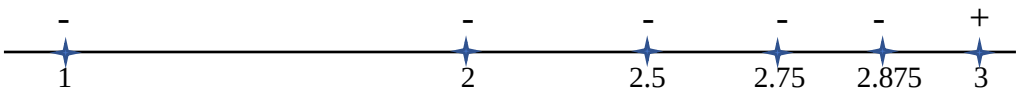
$$n > \frac{2}{\log_{10} 2} = 6.6$$

$$n \cong 7$$

H.W

1. Find an approximation to $\sqrt[3]{25}$ correct to within 10^{-1}, $x \in [1,3]$.

Solution:



n	A	b	p	f(p)
1	1	3	2	-17
2	2	3	2.5	-9.375
3	2.5	3	2.75	-4.203
4	2.75	3	2.875	-1.236
5	2.875	3	2.938	0.36

2. Find an approximation to $\sqrt{3}$ correct to within 10^{-1}, $x \in [1,2]$.

Solution:

n	A	b	p	f(p)
1	1	2	1.5	-0.75
2	1.5	2	1.75	0.063
3	1.5	1.75	1.625	-0.359
4	1.625	1.75	1.688	-0.151

3. Find the root $f(x) = x^3 + 4x^2 - 10$, $x \in [1,2]$ by using bisection method with error $\varepsilon = 10^{-3}$.

4. Find the approximation value of the root of

a. $f(x) = xe^x - 1$ with $\varepsilon = 0.005$

b. $f(x) = e^x - 2$, $I = [0,1]$, $\varepsilon = 0.01$

Algorithm (Bisection Method)

```
F=@(x)cos(x)-exp(x)*x^2+2;
a=0; b=1; imax=35; tol=0.0005;
Fa=F(a); Fb=F(b);
disp ('Bisection Method:');
if Fa*Fb>0
   disp('Error:The function has the same sing at points a and b.')
else
   disp('iteration    a          b          (xNS)       f(xNS)     Tolerance')
   for i=1:imax
      xNS=(a+b)/2;
      toli=(b-a)/2;
      FxNS=F(xNS);
      fprintf('%3i   %11.6f   %11.6f   %11.6f   %11.6f   %11.6f\n',i,a,b,xNS,FxNS,toli)
      if FxNS == 0
        fprintf('An exact solution x=%11.6f was found',xNS)
        break
      end
      if toli<tol
        break
      end
      if i == imax
        fprintf('Solution was not obtained in %i iterations',imax)
        break
      end
      if F(a)*FxNS<0
        b=xNS;
      else
        a=xNS
      end
   end
end
```